

19.1: Language of motion:

Scalar: A quantity which only has a magnitude
Vector: A quantity which has both magnitude & direction
Distance: Total ground covered independent of direction - scalar.
Position: Distance from a fixed distance - vector.
Displacement: Distance from any fixed position - vector.
Use diagrams & graphs to sketch problems.

Notation:

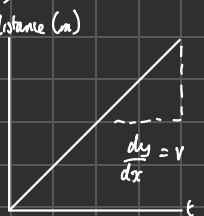
s, h, x, y, z: Position
t - time from a starting instant.
u, v - velocities
a - acceleration.

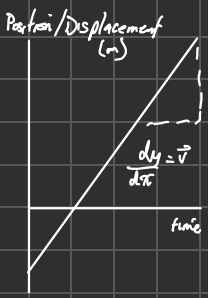
SI Units:

Distance - metre (m)
Time - Seconds (s)
Mass - kilograms (kg)
Speed - metres per second (ms^{-1})

19.2: Speed & Velocity:

Speed: Rate of change of distance - scalar.
Velocity - Rate of change of displacement - vector.
Velocity can be negative (it has a dir^{n}), but speed cannot.



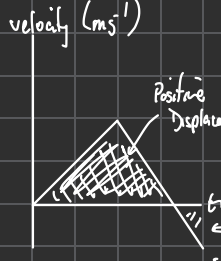


$\bar{v} = \frac{\text{total distance}}{\text{time taken}}$, $\vec{v}_{avg} = \frac{\text{total displacement}}{\text{time taken}}$

Velocity/speed can be found at an instant using trivial techniques.

19.3: Acceleration:

$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$



$\frac{dv}{dt} = a$, gradient of graph is acceleration

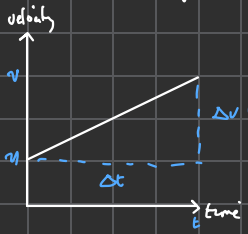
$\int v\,dt = s$, area under graph is displacement.

Positive Acceleration means velocity is increasing
Negative Acceleration means a decreasing velocity.
Deceleration = Negative Acceleration

19.4: Using Areas to find distances & displacements:

- Area under a speed-time graph is distance travelled.
- Area under a velocity-time graph is displacement.
- To find areas, find the equation to $S(t)$, or use $V(t)$ and integrate. Can also make pretty shapes ☺.

19.5: SUVAT Equations:



Acceleration in free fall is $9.8 ms^{-2}$ (unless said otherwise), due to gravity. Acts vertically downwards.

1: $v = u + at$

$\Rightarrow \frac{\Delta v}{\Delta t} = a = \frac{v-u}{t} \Rightarrow v = u + at$

2: $s = ut + \frac{1}{2}at^2$

① $\Rightarrow A = ut + \frac{(v-u)t}{2} = ut + \frac{(at)t}{2} = ut + \frac{1}{2}at^2$

② $\int v\,dt = \int u + at\,dt = s = ut + \frac{1}{2}at^2$

3: $s = \frac{(u+v)t}{2}$

$\Rightarrow A_{trap} = \frac{(u+v)t}{2} = s.$

4: $v^2 = u^2 + 2as$

$\Rightarrow s = \frac{(u+v)t}{2} = \frac{(u+v)(v-u)}{2a}$ (①) $= \frac{v^2-u^2}{2a} \Rightarrow v^2 - u^2 = 2as$

$\Rightarrow v^2 = u^2 + 2as$

When there are more complex problems, if an object starts after time t, it's time is $(t-a)$.
Same principle with displacement: